Ian Wang

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Technical Skills

Languages: Python, JavaScript, TypeScript, Java, C, SAS, SQL
Web & Mobile: React, React Native, Node.js, Express.js, Firebase
Cloud & DevOps: AWS, Terraform, Docker, Git, CI/CD, Azure DevOps

Experience

Catalina Marketing Chicago, IL

Catalina Marketing specializes in shopper intelligence and personalized marketing solutions for CPG brands, retailers and agencies.

Data Scientist Oct. 2018 – Sep. 2023

- Developed shopper-targeting models based on requirements gathered from internal business units, directly supporting marketing campaigns associated with approximately $15M in annual revenue.
- Collaborated with cross-functional teams to deploy propensity models through an Agile workflow in Azure DevOps, identifying high-potential buyers from over 100M customers.
- Streamlined model development in Azure Databricks by restructuring resource-intensive data preparation processes, increasing production capacity by 20%.
- Integrated third-party demographic and purchase data to enhance predictive model performance, expanding target audience reach beyond the in-store network to support digital campaign scaling.
- Refactored legacy SAS production code into modular Python scripts, executing backtesting plans to validate model performance.
- Automated campaign reporting by integrating external vendor data via RESTful API, improving campaign management efficiency.

Senior Analyst Apr. 2016 – Sep. 2018

- Conducted post-campaign analyses to evaluate marketing measurement, providing lift and ROI insights to senior team members for client reporting and retention.
- Delivered shopper insights and ad hoc analyses using SAS and SQL, directly supporting sales team efforts in client acquisition and upsells with data-driven findings.
- Supported team leadership and sales in translating client needs and business objectives into analytical solutions and presentations, driving business development through data insights.

Education

Northeastern University Vancouver, BC
M.S. in Computer Science *Apr. 2026*

Rutgers University New Brunswick, NJ
M.S. in Statistics *May 2015*

Projects

Cloud-Native E-Commerce Architecture Study | *AWS, Terraform, Docker, Locust* Nov. 2025

- Built and deployed a containerized e-commerce system on AWS; conducted comparative performance analysis of Monolithic vs. Microservices architectures under simulated loads of 300+ concurrent users via Locust.
- Analyzed architectural tradeoffs revealing microservices achieved higher throughput through independent scaling while monolith showed lower P95 latency at a moderate scale, demonstrating architecture choice depends on scale requirements not absolute performance.
- Identified database as the primary bottleneck preventing performance improvements through horizontal scaling.

E-Commerce Web Application | *JavaScript, MongoDB, Express.js, React, Node.js* Nov. 2024

- Built a full-stack e-commerce web application using the MERN stack with core CRUD operations and JSON Web Token (JWT) for user authentication.
- Leveraged AI coding tools to accelerate code generation, debugging, and refactoring for improved code quality and maintainability.
- Achieved 80%+ test coverage using Jest for component and API testing.